Google Play Integrity — Android Integration Guide Overview Every protected API call (game completion, daily rewards, achievements, wallet) requires a valid integrity token. The f bw is:

App → GET nonce from backend → Pass to Play Integrity API → Get token → Send with request Step 1 — Add Dependency In app/build.gradle:

dependencies { implementation 'com.google.android.play:integrity:1.3.0' } Step 2 — Create IntegrityManager (once at app start)

import com.google.android.play.core.integrity.IntegrityManagerFactory import com.google.android.play.core.integrity.IntegrityTokenRequest

class IntegrityHelper(private val context: Context) {

```
private val integrityManager = IntegrityManagerFactory.create(context)

suspend fun getIntegrityToken(): String {
    // Step 1: Get nonce from your backend
    val nonce = fetchNonceFromBackend()

    // Step 2: Request token from Google
    val request = IntegrityTokenRequest.newBuilder()
        .setNonce(nonce)
        .build()

    return suspendCoroutine { continuation ->
        integrityManager.requestIntegrityToken(request)
            .addOnSuccessListener { response ->
                continuation.resume(response.token())
            }
            .addOnFailureListener { exception ->
                continuation.resumeWithException(exception)
            }
    }
}

private suspend fun fetchNonceFromBackend(): String {
    // POST /api/integrity/challenge
    val response = apiService.getIntegrityChallenge(
        authHeader = "Bearer $userToken"
    )
    return response.data.nonce
}
```

} Step 3 — Send Token With Every Protected Request Add x-integrity-token and x-platform headers to all protected API calls:

// In your API interceptor (OkHttp/Retrof t) class IntegrityInterceptor(private val integrityHelper: IntegrityHelper) : Interceptor {

integrityHelper: IntegrityHelper): Interceptor {

```
override fun intercept(chain: Interceptor.Chain): Response {
    val originalRequest = chain.request()

    // Only add token to protected endpoints
    val protectedPaths = listOf(
        "/api/daily-rewards",
        "/api/games",
        "/api/achievements",
        "/api/wallet"
    )

    val isProtected = protectedPaths.any {
        originalRequest.url.encodedPath.contains(it)
    }

    if (!isProtected) return chain.proceed(originalRequest)

    // Get token (runs on IO thread)
    val token = runBlocking {
        try {
            integrityHelper.getIntegrityToken()
        } catch (e: Exception) {
            return@runBlocking null
        }
    }

    val newRequest = originalRequest.newBuilder()
        .addHeader("x-platform", "android")
        .apply {
            if (token != null) addHeader("x-integrity-token", token)
        }
        .build()

    return chain.proceed(newRequest)
}
```

} Step 4 — Register Interceptor in Retrofit

val okHttpClient = OkHttpClient.Builder()
 addInterceptor(AuthInterceptor(tokenManager))
 addInterceptor(IntegrityInterceptor(integrityHelper)) // ← add this build()

val retrofit = Retrofit.Builder() baseUrl("https://your-api.com") client(okHttpClient)
 addConverterFactory(GsonConverterFactory.create()) build() Step 5 — Backend API Reference Get Nonce

POST /api/integrity/challenge Authorization: Bearer

Response: { "success": true, "data": { "nonce": "a3f9c2...(64 hex chars)", "packageName":

"com.jackson.rewards", "timestamp": 1711186800000 } Headers required on protected requests

Authorization: Bearer x-integrity-token: x-platform: android Content-Type: application/json Important Notes Rule Detail Nonce is single-use Request a fresh nonce for every protected action — never reuse Nonce expires Valid for 5 minutes only Package name Must be com.jackson.rewards — same as in Play Console iOS Send x-platform: ios — no integrity token needed for iOS Missing token Backend returns 401 INTEGRITY_TOKEN_REQUIRED Invalid token Backend returns 403 INTEGRITY_VERIFICATION_FAILED Error Handling

try { val token = integrityHelper.getIntegrityToken() // proceed with API call } catch (e: IntegrityServiceException) { when (e.errorCode) { IntegrityErrorCode.NO_ERROR -> { /* success */ } IntegrityErrorCode.PLAY_STORE_NOT_FOUND -> { // Device doesn't have Play Store — show message showError("Google Play Store is required") } IntegrityErrorCode.APP_NOT_INSTALLED -> { // App not installed via Play Store showError("Please install the app from Play Store") } IntegrityErrorCode.NETWORK_ERROR -> { // Retry retryRequest() } else -> showError("Integrity check failed") }}Test Checklist POST /api/integrity/challenge returns a nonce Token generated successfully on real device Token sent in x-integrity-token header Protected API calls succeed with token API returns 401 when token is missing Nonce cannot be reused (second request with same nonce returns 403)